Xifan Yu

Contact Information

Email: xifan.yu@yale.edu Tel: (773) 690-2203
Homepage: xifanyu.github.io

Research Interests

I am broadly interested in theoretical computer science, including graph theory, average-case complexity, spectral methods, Sum-of-Squares algorithms, and high-dimensional statistics. Recently I am interested in finite free probability and the theory of language generation.

Education

Yale University *Sept 2021 - Present*
Ph.D. in Computer Science (expected June 2027)

University of Chicago *Sept 2017 - June 2021*
M.S. in Computer Science (June 2021)
B.S. in Computer Science and Mathematics, *summa cum laude* (June 2021)

Teaching Experience

Yale University, Department of Computer Science *Sept 2022 - May 2023*
Teaching Fellow *New Haven, CT*

- CPSC 365 Algorithms, Dr. Dylan McKay, Spring 2023
- CPSC 202 Mathematical Tools for CS, Dr. Dylan McKay, Fall 2022

University of Chicago, Department of Computer Science *Oct 2020 - Mar 2021*
Teaching Assistant *Chicago, IL*

- CMSC 27230 Honors Theory of Algorithms, Prof. László Babai, Winter 2021
- CMSC 37115 Intro to Mathematical Reasoning, Prof. László Babai, Fall 2020

University of Chicago, Department of Computer Science *April 2019 - June 2020*
Grader *Chicago, IL*

- CMSC 27500 Graph Theory, Prof. Ketan Mulmuley, Spring 2020
- CMSC 37000 Algorithms, Prof. Yury Makarychev, Winter 2020
- CMSC 27530 Honors Graph Theory, Prof. László Babai, Spring 2019

Industry Experience

Horizon Robotics Inc. *June 2019 - Sept 2019*
Software Engineer *Cupertino, CA*

Publications and Preprints

Stable algorithms lower bounds for estimation from MMSE discontinuities. Xifan Yu, Ilias Zadik. *To Appear in Conference of Learning Theory (COLT)*, 2026.

Differentially private language generation in the limit. Anay Mehrotra, Grigoris Velegkas, Xifan Yu, Felix Zhou. *To Appear in Conference of Learning Theory (COLT)*, 2026.

Language generation with infinite contamination. Anay Mehrotra, Grigoris Velegkas, Xifan Yu, Felix Zhou. *To Appear in Conference of Learning Theory (COLT)*, 2026.

Counting stars is constant-degree optimal for detecting any planted subgraph. Xifan Yu, Ilias Zadik, Peiyuan Zhang. *Mathematical Statistics and Learning*, 2025.

Statistical inference of a ranked community in a directed graph. Dmitriy Kunisky, Daniel A. Spielman, Alexander S. Wein, Xifan Yu. *Symposium on Theory of Computing (STOC)*, 2025.

Computational hardness of detecting graph lifts and certifying lift-monotone properties of random regular graphs. Dmitriy Kunisky, Xifan Yu. *Symposium on Foundations of Computer Science (FOCS)*, 2024.

Counting stars is constant-degree optimal for detecting any planted subgraph (Extended Abstract). Xifan Yu, Ilias Zadik, Peiyuan Zhang. *Conference on Learning Theory (COLT)*, 2024.

A degree 4 sum-of-squares lower bound for the clique number of the Paley graph. Dmitriy Kunisky, Xifan Yu. *Computational Complexity Conference (CCC)*, 2023.

Honors and Awards

- **Phi Beta Kappa, Inducted** *June 2020*
  University of Chicago
- **Student Marshall** *June 2020*
  University of Chicago
- **Dean's List** *2017-2018, 2018-2019, 2019-2020*
  University of Chicago

Programming Competition Awards

- **Top 1000 in Round 2** *May 2020*
  Google Code Jam 2020
- **North America Finalist** *Feb 2020*
  International Collegiate Programming Contest's inaugural North America Championship
- **Top 500 in Round 2** *July 2019*
  Facebook Hacker Cup 2019
- **World Finalist** *April 2019*
  43rd Annual World Finals of the International Collegiate Programming Contest

References

Daniel A. Spielman
Sterling Professor of Computer Science
Professor of Statistics and Data Science and of Mathematics
Yale University
daniel.spielman@yale.edu

Dmitriy Kunisky
Assistant Professor of Applied Mathematics and Statistics
Johns Hopkins University
kunisky@jhu.edu

Ilias Zadik
Assistant Professor of Statistics and Data Science
Yale University
ilias.zadik@yale.edu